

Swayam Gupta

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Professional Summary

Backend-focused software engineering student passionate about distributed systems, networking, concurrency, and cloud infrastructure. Experienced building scalable applications using Go, Python, Node.js, PostgreSQL, Redis, and AWS, with a strong foundation in operating systems, computer networks, and data structures.

Technical Skills

Languages: Go (Golang), Python, C/C++, TypeScript, JavaScript, SQL

Backend: Node.js, Express.js, FastAPI, PostgreSQL, MongoDB, Redis, Supabase, Prisma, REST APIs, WebSockets

AI & Systems: LangGraph, Gemini, RAG, ChromaDB, Sentence Transformers, Linux, AWS, Docker, TCP/UDP Networking, Git

Developer Tools: GitHub Actions, Claude Code, OpenRouter, Codex

Projects

RescueNet | Go, mDNS, TCP/UDP, Peer-to-Peer Networking

[GitHub](#)

- Engineered an offline-first peer-to-peer communication layer using **Go** and **goroutines**, enabling nearby devices to communicate without internet connectivity.
- Implemented **mDNS-based service discovery** for automatic peer discovery and decentralized network formation without relying on centralized servers.
- Designed concurrent message handling, connection management, and reliable packet delivery using Go's concurrency primitives for fault-tolerant communication.

Unbuilt | Python, LangGraph, Gemini, FastAPI, AWS, SQS, Supabase, Docker

[GitHub](#)

- Built a **multi-agent LangGraph system** using Gemini to analyze Reddit/Hacker News demand signals and generate personalized, evidence-backed project ideas.
- Designed an **asynchronous AWS architecture** using API Gateway, Lambda, SQS, and Supabase to decouple long-running AI workflows from HTTP requests.
- Containerized the backend with **Docker** and **Amazon ECR** and implemented **GitHub Actions CI/CD** for automated testing, image publishing, and Lambda deployment.

SimplyRAG | Python, FastAPI, ChromaDB, Sentence Transformers, BM25, RAGAS

[GitHub](#)

- Built a production-oriented **Retrieval-Augmented Generation (RAG)** pipeline for querying information from uploaded PDF documents using semantic and lexical retrieval.
- Implemented **hybrid retrieval** combining ChromaDB vector search with BM25 keyword retrieval, followed by **Cross-Encoder reranking** to improve document relevance.
- Added **RAGAS evaluation** and automated benchmarking to measure retrieval and answer quality across test queries, with a FastAPI interface for serving the pipeline.

Achievements & Extracurriculars

- Algorithmic Problem Solving:** Solved **400+** data structures and algorithms problems on **LeetCode** using **C++**, covering graphs, trees, dynamic programming, greedy algorithms, and advanced data structures.
- Technical Mentorship:** Conducted offline sessions on **C++** and data structures for peers and junior students.
- Certifications:** Completed *Mastering Data Structures and Algorithms (C/C++)* and *Graph Data Structures for Competitive Programming*.

Education

Bhagwan Parshuram Institute of Technology

2023 – 2027

Bachelor of Technology, Computer Science and Engineering

Rohini, Delhi

CGPA: **8.50**

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Cloud Computing, Database Management Systems, Software Engineering

Delhi Public School

2011 – 2023

Higher Secondary (90.4%) | Secondary (88%)

Jammu, J&K